Forecasting with time-based data

Companion to the three short videos. Example: orders per hour at a small café.

The words

Regression	predict a number (classification predicts a category).
Forecast	a number for a future period.
Feature	an input the model uses, such as weekday, hour, holiday.
Leakage	using a value you would not have yet when you forecast.

Test over time

Train on the earlier rows. Test on the latest rows. The model must never see the future while it learns.

Never split rows at random: the model would train on the neighbours of the very hours it is tested on, and the score looks better than a real forecast would.

Features: the test

At the moment I make the forecast, do I know this value?

Yes (weekday, hour, holiday, week number): may use. No (the previous hour of the day you are forecasting): may not use.

WAPE: how wrong is the forecast?

WAPE = total absolute error / total actual

Hour	Actual	Forecast	Abs. error
A	40	30	10
B	10	20	10
C	50	45	5
Total	**100**		**25**

WAPE = 25 / 100 = 25%. Lower is better; 0% is perfect.

Baseline: the number to beat

Repeat the last week you know, hour for hour. It takes a minute. A model that cannot beat it on the held-out period is not useful.

Reading the result

A forecast is an expectation, not a promise. Hourly counts are noisy, so a WAPE around 15-20% can be normal. Check the error in the busy hours: they matter most.

Try it yourself: the café data (cafe_orders.csv) and the numbers to expect

Columns: date, hour, orders, holiday. 12 weeks, hourly, 07:00-18:00. Train on weeks 1-10, test on weeks 11-12 (the last two weeks).

Baseline: repeat week 10, hour for hour	WAPE = 22.9%
Weekday-hour average: mean of weeks 1-10	WAPE = 17.5%

If your numbers match, your split, baseline and WAPE are right. Any model you build should be judged the same way: on the held-out weeks, against the baseline.

The café data is synthetic and made for this introduction.